MATH 202 Your name: _____

Learning target T1, version 1

Find the Taylor polynomials of a given function about a given center a.

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1. Suppose that $f(x)$ is some function, and that we know the following information:

$$f(-1) = 3,\ f'(-1) = \frac{1}{2},\ f''(-1) = -1,\ f'''(-1) = 0,\ f^{(4)}(-1) = \frac{1}{5}$$

   Determine $T_4(x)$, the degree 4 Taylor polynomial approximation to $y = f(x)$ centered at $a = -1$.

2. Now suppose that $g(x) = \dfrac{1}{x} = x^{-1}$. Find the degree 2 Taylor polynomial approximation, $T_2(x)$, to $g(x)$, centered at $a = 3$. Show your work!

MATH 202 Your name: _____

Learning target T1, version 2

Find the Taylor polynomials of a given function about a given center a.

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1. Suppose that $f(x)$ is some function, and that we know the following information:

$$f(-3)=2,\ f'(-3)=-4,\ f''(-3)=1, f'''(-3)=0,\ f^{(4)}(-3)=5,\ f^{(5)}(-3)=-2$$

   Determine $T_4(x)$, the degree 4 Taylor polynomial approximation to $y=f(x)$ centered at $a=-3$.

2. Now suppose that $g(x)=\sqrt{x}=x^{1/2}$. Find the degree 2 Taylor polynomial approximation, $T_2(x)$, to $g(x)$, centered at $a=64$. Show your work!

3. (Bonus) Find a good rational approximation to, uhhhhhh, $\sqrt{67}$, waving my hands up and down.